Synopsis

Heart Disease Prediction using Machine Learning Algorithms

Problem Id: 13

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- **Problem Description:**

One of the major health issues that cause the highest number of deaths worldwide is heart disease. It is, therefore, crucial to prevent and detect heart disease at an early stage. The purpose of the project is to create a machine learning model to predict the occurrence of heart disease based on the data provided by patients. The model will be trained to process different health-related parameters, such as age, blood pressure, cholesterol level, and so on, to check whether the patient is suffering from heart disease or not.

Using machine learning, the model will be able to help medical professionals take faster and more accurate decisions. The project is based on the development of a machine learning model that includes data preprocessing, model development, and evaluation. The final output will be the classification of patients based on the occurrence of heart disease, either yes or no. This project shows the practical usage of AI in the real world.

- **Steps of Implementation:**

1. Data Collection

The dataset for heart disease prediction is collected from a reliable source such as the UCI Machine Learning Repository. It contains clinical features like age, sex, blood pressure, cholesterol, etc., along with the target variable indicating the presence or absence of heart disease.

2. Data Preprocessing

The collected data is cleaned by handling missing values using mean or median techniques. Categorical variables are converted into numerical form using encoding methods. The dataset is also normalized/standardized to ensure that all features contribute equally to the model.

3. Model Selection

Suitable machine learning algorithms are selected for classification, including Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). These models are chosen because they are effective for binary classification problems and provide good performance on structured data.

4. Training and Validation

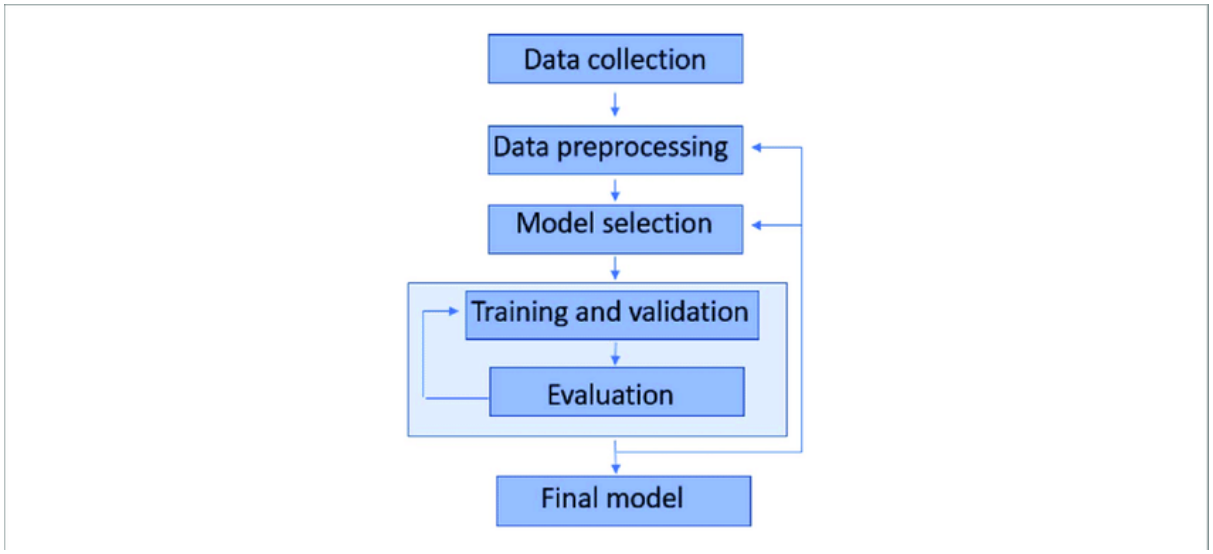
The dataset is divided into training and testing sets. The models are trained using the training data, and validation is performed using the test data to evaluate how well the models generalize to unseen data.

5. Evaluation

The performance of each model is evaluated using metrics such as accuracy, precision, recall, and F1-score. Additional tools like confusion matrix and ROC curve are used to visualize and compare model performance.

6. Final Model

After comparing all models, the best-performing model is selected based on evaluation metrics. This final model is used to predict whether a patient has heart disease (1) or not (0).



- **Expected Modules/ Libraries to be used :**

  NumPy – Used for numerical computations
  Pandas – Used for data handling and preprocessing
  Matplotlib – Used for data visualization
  Seaborn – Used for advanced visualizations
  sklearn – Used for ML models and evaluation

- **Expected Evaluation metrics and Plots to be used:**

In this project, the performance of machine learning models will be evaluated using basic and commonly used evaluation metrics.

- **Accuracy –** It measures the overall correctness of the model by calculating the ratio of correctly predicted instances to total instances.
- **Confusion Matrix –** It is used to visualize the performance of the classification model by showing correct and incorrect predictions.

- **Error Rate –** It represents the percentage of incorrect predictions made by the model.
- **Scatter Plot** – Used to visualize relationships between features (especially useful for understanding data patterns in Linear Regression).
- **Bar Graph** – Used to compare the performance (accuracy) of different models like KNN and Logistic Regression.
- **Histogram** – Used to understand the distribution of different features in the dataset.

- **Required topics from the subject:**
    1. **Data Preprocessing** – Cleaning data, handling missing values, and encoding.
    2. **Feature Scaling** – Normalization and standardization for better model performance.
    3. **Supervised Learning** – Learning from labeled data for classification.
    4. **Linear Regression** – Basic algorithm to understand relationships in data.
    5. **K-Nearest Neighbors (KNN)** – Distance-based classification algorithm.
    6. **Model Evaluation** – Accuracy and confusion matrix to evaluate models.
    7. **File Handling** – Reading dataset from CSV files.

- **Books and Link Sources:**
    1. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*
    2. *Introduction to Machine Learning with Python*
    3. UCI Machine Learning Repository (Dataset source)
    4. Scikit-learn Documentation
    5. Python Official Documentation
    6. Kaggle